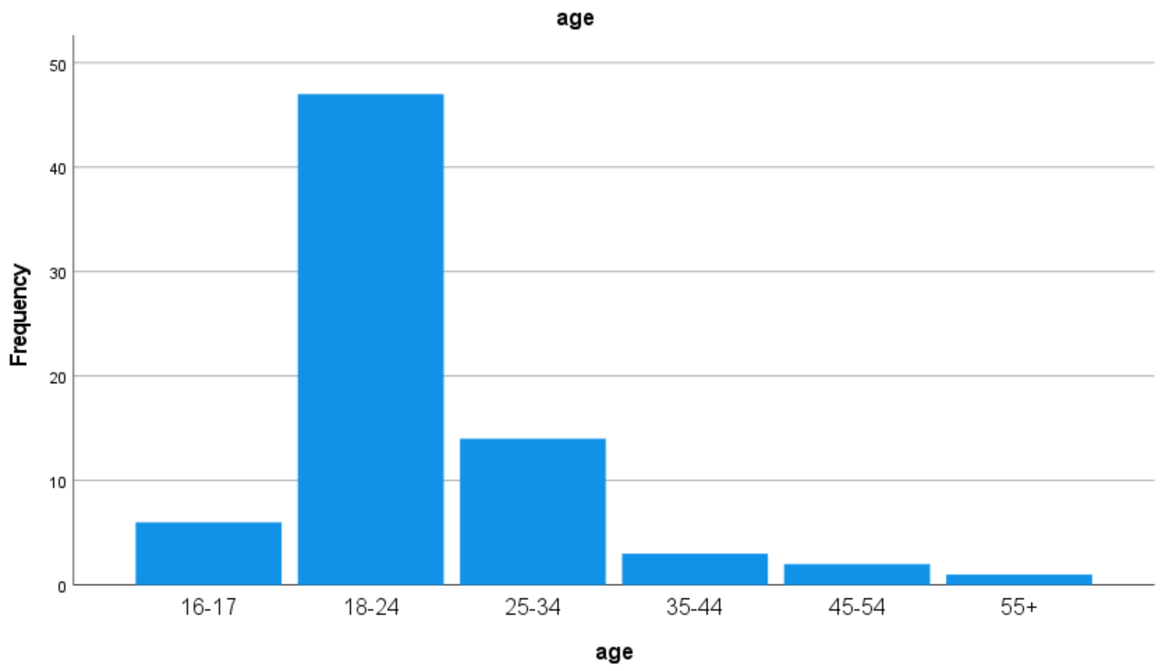
Data Analysis

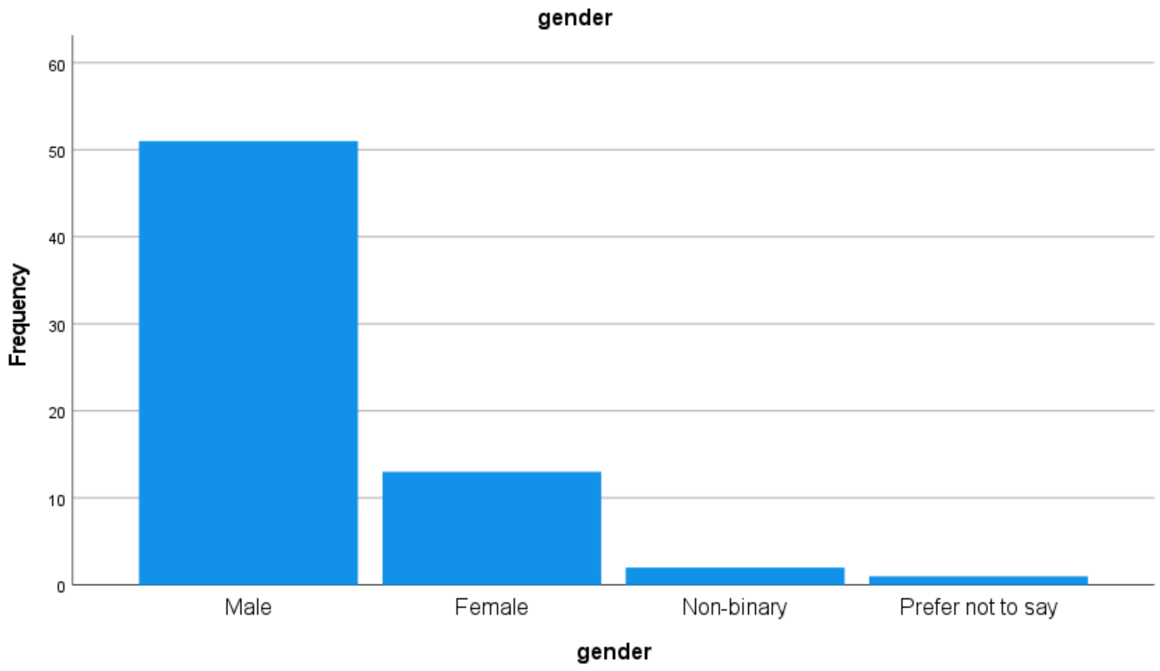
09. June 2026

Descriptive statistics

- Missing values -> None
  - Empty gender values filled with 999(unknown)
  - Empty age values filled with age group 16-17
- Wrong values -> None
- Duplications -> Fixed in data cleaning
- Exclusions -> 5 removed during cleaning
  - Reason: Did not finish the full procedure

Participant characteristics





Normality

Tests of Normality

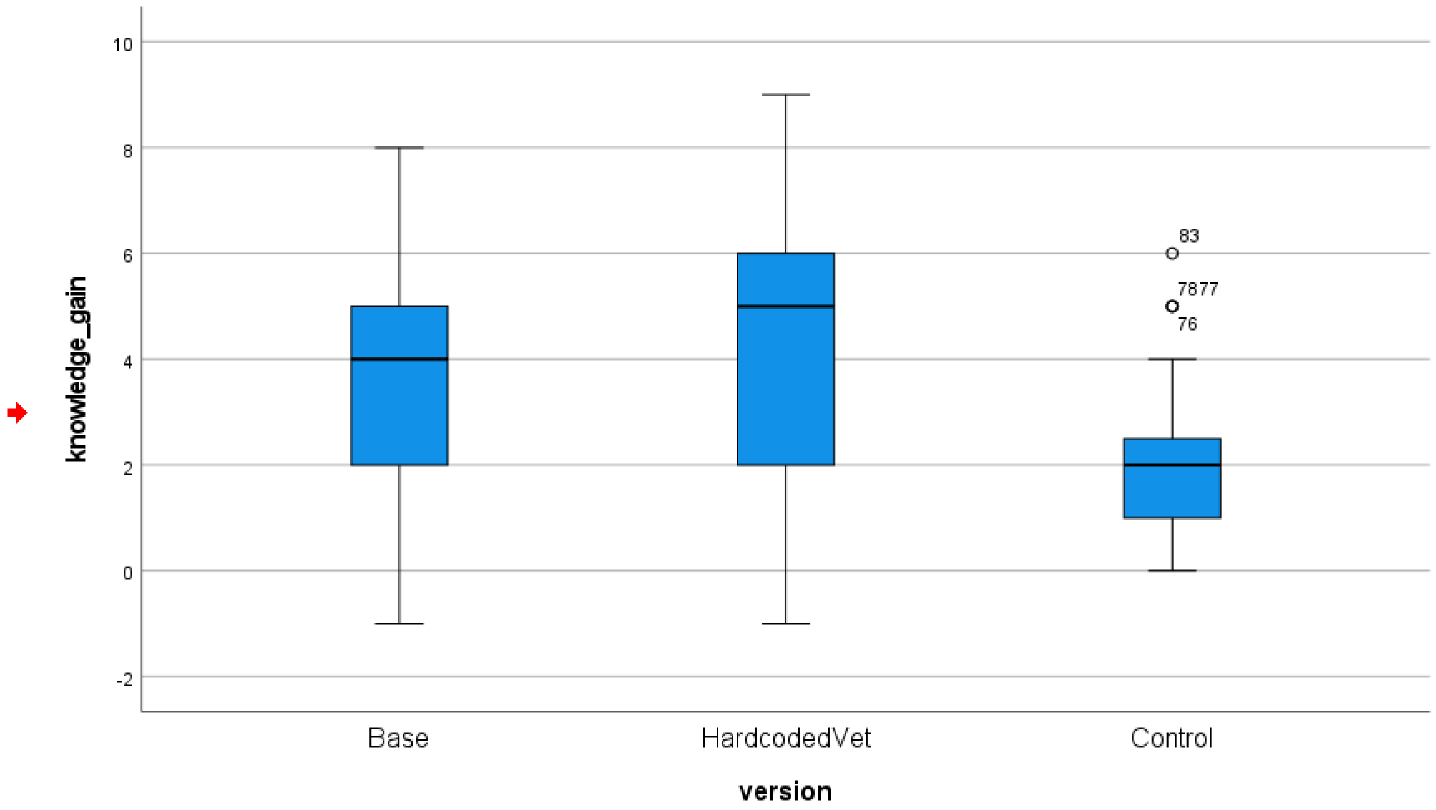
	version	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
pre_knowledge_score	Base	,150	25	,150	,906	25	,025
	HardcodedVet	,171	21	,110	,927	21	,117
	Control	,145	27	,149	,932	27	,076
post_knowledge_score	Base	,175	25	,046	,949	25	,240
	HardcodedVet	,207	21	,020	,919	21	,081
	Control	,198	27	,008	,906	27	,018
knowledge_gain	Base	,151	25	,143	,955	25	,327
	HardcodedVet	,133	21	,200*	,971	21	,749
	Control	,297	27	<,001	,872	27	,003
time_played_s	Base	,089	25	,200*	,987	25	,979
	HardcodedVet	,153	21	,200*	,954	21	,399
	Control	,088	27	,200*	,979	27	,845
engagement_score	Base	,087	25	,200*	,983	25	,934
	HardcodedVet	,138	21	,200*	,967	21	,660
	Control	,142	27	,174	,974	27	,718

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Outliers

No outliers in all variables except knowledge gain



Remove outliers?
No. The entries are valid.
These seem more like high performers.
(Possibility of cheating?)

Baseline equivalence

- One-way ANOVA on pre-knowledge score

Descriptives

pre_knowledge_score

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound	95% Confidence Interval for Mean Upper Bound	Minimum	Maximum
Base	25	3,00	2,273	,455	2,06	3,94	0	7
HardcodedVet	21	2,52	1,887	,412	1,66	3,38	0	6
Control	27	2,93	1,880	,362	2,18	3,67	0	6
Total	73	2,84	2,007	,235	2,37	3,30	0	7

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
pre_knowledge_score	Based on Mean	,621	2	70	,541
	Based on Median	,637	2	70	,532
	Based on Median and with adjusted df	,637	2	68,829	,532
	Based on trimmed mean	,688	2	70	,506

ANOVA

pre_knowledge_score

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	2,937	2	1,469	,358	,700
Within Groups	287,090	70	4,101		
Total	290,027	72			

Question: Do the groups differ significantly in ther initial knowledge score?

Answer: No

Knowledge variable

Repeated measures GLM

- Within:
  - time (2 levels)
- Variables:
  - pre score
  - post score
- Between:
  - version

Questions:
(main effect) Did participants learn overall?
(interaction) Did learning differ between conditions?

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
time	Pillai's Trace	,706	168,260[b]	1,000	70,000	<,001
	Wilks' Lambda	,294	168,260[b]	1,000	70,000	<,001
	Hotelling's Trace	2,404	168,260[b]	1,000	70,000	<,001
	Roy's Largest Root	2,404	168,260[b]	1,000	70,000	<,001
time * version	Pillai's Trace	,140	5,685[b]	2,000	70,000	,005
	Wilks' Lambda	,860	5,685[b]	2,000	70,000	,005
	Hotelling's Trace	,162	5,685[b]	2,000	70,000	,005
	Roy's Largest Root	,162	5,685[b]	2,000	70,000	,005

a. Design: Intercept + version
Within Subjects Design: time

b. Exact statistic

With contrasts

/CONTRAST(version)=Simple

- It is doing simple contrasts on the **between-subjects factor** version for the **Averaged variable MEASURE_1**, i.e. the mean of pre and post together. NOT ON THE GAIN.
- a **between-subjects effect on the overall average level**, not on the *change* from pre to post.
- averaging pre and post dilutes the difference

Contrast Results (K Matrix)

version Simple Contrast[a]			Averaged Variable MEASURE_1
Level 1 vs. Level 3	Contrast Estimate		,743
	Hypothesized Value		0
	Difference (Estimate - Hypothesized)		,743
	Std. Error		,384
	Sig.		,057
	95% Confidence Interval for Difference	Lower Bound	-,024
		Upper Bound	1,510
Level 2 vs. Level 3	Contrast Estimate		,653
	Hypothesized Value		0
	Difference (Estimate - Hypothesized)		,653
	Std. Error		,403
	Sig.		,109
	95% Confidence Interval for Difference	Lower Bound	-,150
		Upper Bound	1,457

a. Reference category = 3

1 = Base
2 = Hardcoded
3 = Control

Engagement variable

One-way ANOVA

- On GEQ scores

Question: Does engagement differ?

Answer: Nothing significant!

Descriptives

engagement_score

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Base	25	48,72	6,624	1,325	45,99	51,45	35	62
HardcodedVet	21	49,81	7,068	1,542	46,59	53,03	35	64
Control	27	46,07	7,076	1,362	43,27	48,87	30	59
Total	73	48,05	7,008	,820	46,42	49,69	30	64

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
engagement_score	Based on Mean	,217	2	70	,805
	Based on Median	,230	2	70	,795
	Based on Median and with adjusted df	,230	2	67,418	,795
	Based on trimmed mean	,214	2	70	,808

ANOVA

engagement_score

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	181,651	2	90,825	1,896	,158
Within Groups	3354,130	70	47,916		
Total	3535,781	72			

ANOVA Effect Sizes[a,b]

		Point Estimate	95% Confidence Interval	
			Lower	Upper
engagement_score	Eta-squared	,051	,000	,161
	Epsilon-squared	,024	-,029	,137
	Omega-squared Fixed-effect	,024	-,028	,135
	Omega-squared Random-effect	,012	-,014	,073

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Time-on-task analysis

One-way anova

- DV: time_played
- Factor: version

Descriptives

time_played_s

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Base	25	617,680	157,4560	31,4912	552,685	682,675	337,0	960,0
HardcodedVet	21	715,571	207,4024	45,2589	621,163	809,980	382,0	1126,0
Control	27	415,926	150,9077	29,0422	356,229	475,623	91,0	699,0
Total	73	571,219	210,4928	24,6363	522,108	620,331	91,0	1126,0

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
time_played_s	Based on Mean	1,615	2	70	,206
	Based on Median	1,008	2	70	,370
	Based on Median and with adjusted df	1,008	2	56,638	,371
	Based on trimmed mean	1,551	2	70	,219

ANOVA

time_played_s

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	1142686,058	2	571343,029	19,534	<,001
Within Groups	2047434,435	70	29249,063		
Total	3190120,493	72			

ANOVA Effect Sizes[a]

		Point Estimate	95% Confidence Interval	
			Lower	Upper
time_played_s	Eta-squared	,358	,173	,490
	Epsilon-squared	,340	,150	,476
	Omega-squared Fixed-effect	,337	,148	,472
	Omega-squared Random-effect	,202	,080	,309

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Time-on-task post hoc

Post Hoc Tests

Multiple Comparisons

Dependent Variable: time_played_s

	(I) version	(J) version	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Bonferroni	Base	HardcodedVet	-97,8914	50,6239	,172	-222,065	26,282
		Control	201,7541*	47,4685	<,001	85,320	318,188
	HardcodedVet	Base	97,8914	50,6239	,172	-26,282	222,065
		Control	299,6455*	49,7605	<,001	177,589	421,702
	Control	Base	-201,7541*	47,4685	<,001	-318,188	-85,320
		HardcodedVet	-299,6455*	49,7605	<,001	-421,702	-177,589
Games-Howell	Base	HardcodedVet	-97,8914	55,1368	,192	-232,529	36,746
		Control	201,7541*	42,8386	<,001	98,235	305,273
	HardcodedVet	Base	97,8914	55,1368	,192	-36,746	232,529
		Control	299,6455*	53,7756	<,001	168,085	431,206
	Control	Base	-201,7541*	42,8386	<,001	-305,273	-98,235
		HardcodedVet	-299,6455*	53,7756	<,001	-431,206	-168,085

*. The mean difference is significant at the 0.05 level.

Correlations & Predictions

What predicts knowledge gain?

Correlations

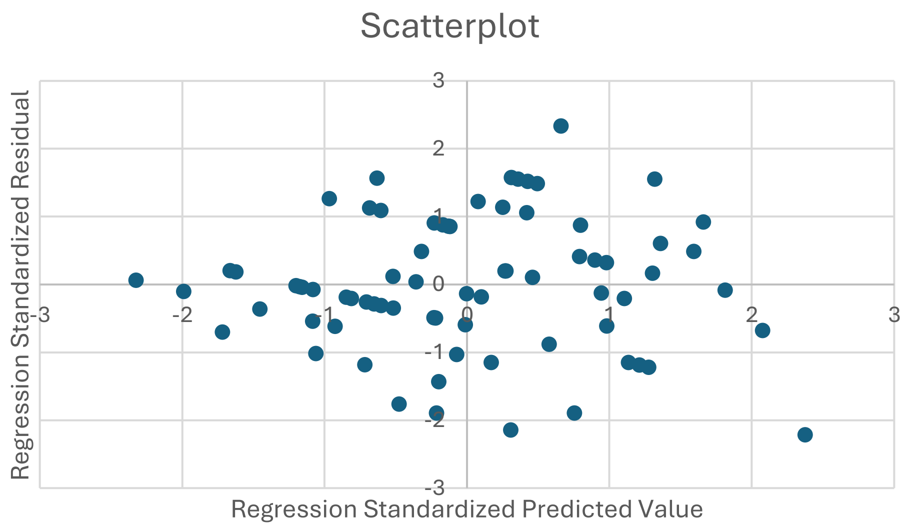
		knowledge_gain	time_played_s	engagement_score
knowledge_gain	Pearson Correlation	1	,414**	,155
	Sig. (2-tailed)		<,001	,190
	N	73	73	73
time_played_s	Pearson Correlation	,414**	1	,047
	Sig. (2-tailed)	<,001		,695
	N	73	73	73
engagement_score	Pearson Correlation	,155	,047	1
	Sig. (2-tailed)	,190	,695	
	N	73	73	73

**. Correlation is significant at the 0.01 level (2-tailed).

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	-,557	2,246		-,248	,805
	version	-,170	,227	-,094	-,750	,456
	time_played_s	,004	,001	,363	2,939	,004
	engagement_score	,040	,037	,120	1,095	,277

a. Dependent Variable: knowledge_gain



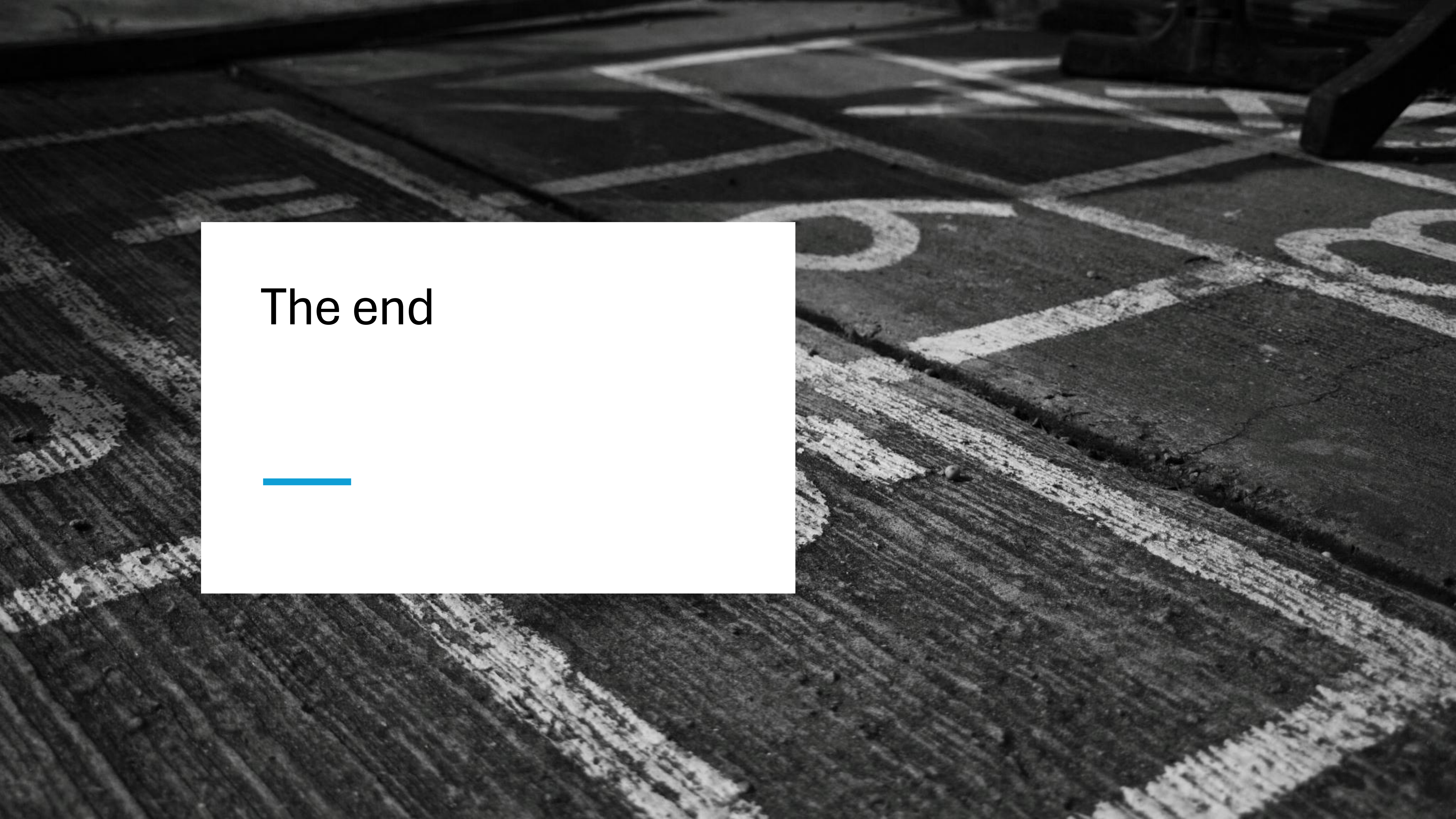
check homoscedasticity (same variance)

Hardcoded Vet

Correlations

Correlations		engagement_score	knowledge_gain	hc_n_unique_nodes
engagement_score	Pearson Correlation	1	-,037	,279
	Sig. (2-tailed)		,873	,221
	N	21	21	21
knowledge_gain	Pearson Correlation	-,037	1	,550**
	Sig. (2-tailed)	,873		,010
	N	21	21	21
hc_n_unique_nodes	Pearson Correlation	,279	,550**	1
	Sig. (2-tailed)	,221	,010	
	N	21	21	21

**. Correlation is significant at the 0.01 level (2-tailed).



The end

Questions